

Session Objectives

At the conclusion of the session, the students should be able to:

1. Understand the movement of the sacrum on the different sacral axes of rotation and the significance of each axis.
2. Understand L5 rules as they relate to sacral torsions
3. Understand all sacral special tests for diagnosis
4. Be able to put information on axes and special tests together to make a sacral diagnosis

Transverse Axes

- Superior
 - Through S2 attachment of dura
 - Primary Respiratory Motion
 - Gross respiratory motion
- Middle
 - Through S2 vertebral body (Also, center of mass of human body)
 - Sacral motion on pelvis
 - Axis of bilateral sacral flexion or extension dysfunctions
- Inferior
 - Through S3
 - Innominate movement relative to sacrum
 - Axis of innominate dysfunctions

Oblique Axis

- From superior SIJ on one side to inferior SIJ on opposite side
- Sacral torsions and rotations
- Named for movement ON axis
- Reference of movement is the anterior sacral body
- Can be torsioned
 - Left on Left (anterior body MUST be going FORWARD)
 - Right on Left (anterior body MUST be going BACKWARD)
 - Imagine a pair of eyes on anterior sacrum... looking left or looking right)
- Can be rotated
 - Same rotation as side of axis is forward, and opposite is backward (don't need to memorize this!)

Torsions vs. Rotations

- Torsion = Twisting
 - Wringing a towel... twisting means turning ends in opposite directions
 - Sacrum turns in opposite direction of L5
- L5 Rules for sacral torsion:
 - Based on assumption that sacrum is in a torsion...
 - Rotation L5 IS side OPPOSITE of sacral 'rotation' (torsion)- by definition
 - **Side bending IS side of oblique axis**
 - L5 Type I dysfunction = Forward/Type II dysfunction = Backward

Physiologic vs. Nonphysiologic

- Normal motion

- Sacrum moves forward and backward (nutation and counternutation) as well as obliquely during normal everyday movements

- Abnormal motion

Physiologic Dysfunctions:

- DYSFUNCTIONAL sacral movement is 'stuck' in a forward direction
- Bilateral sacral flexion dysfunction, forward sacral torsion or rotation

Non-physiologic Dysfunctions:

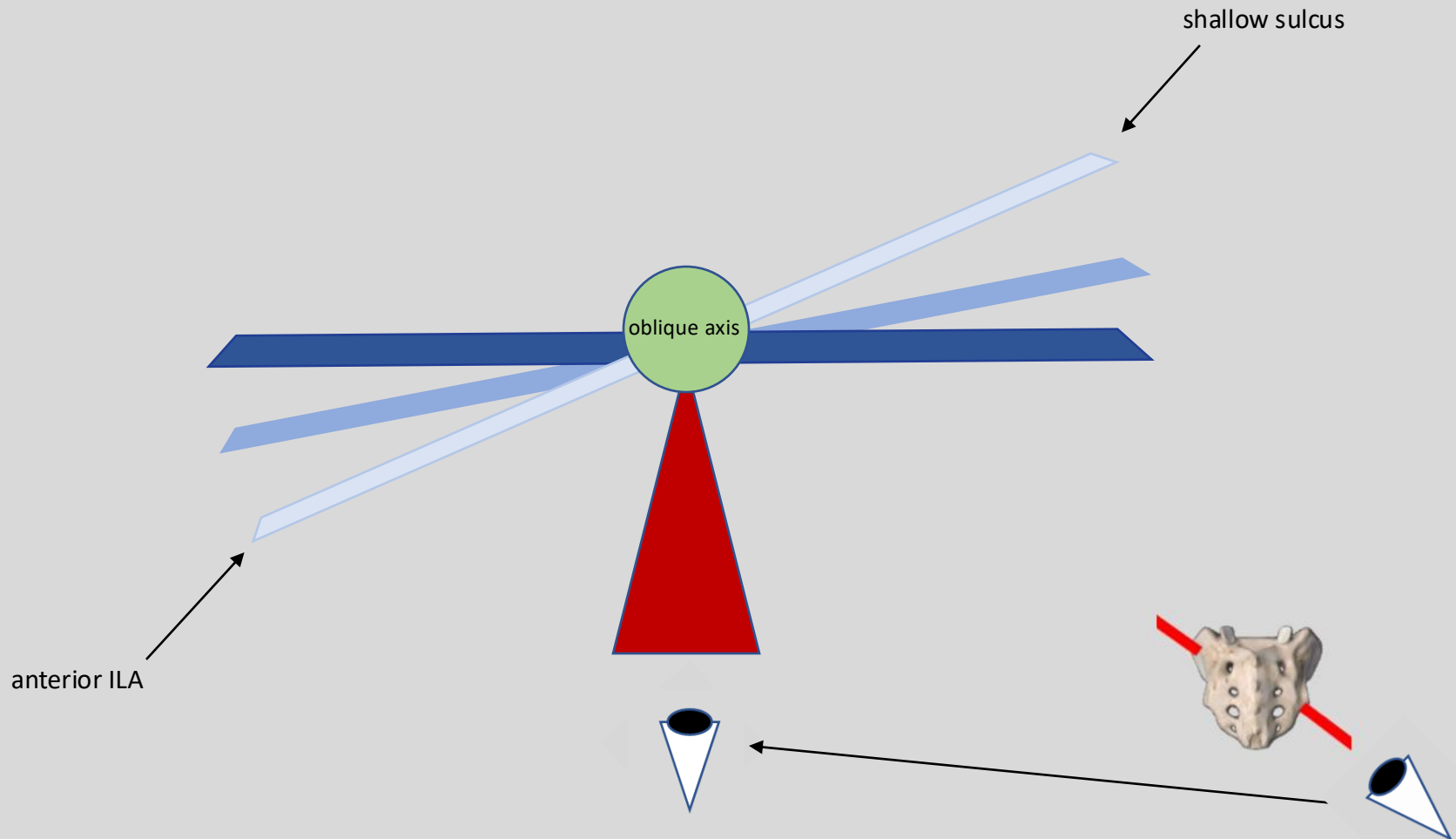
- DYSFUNCTIONAL sacral movement is 'stuck' in a backward direction
- Bilateral sacral extension dysfunction, backward sacral torsion or rotation
- Sacral shears (unilateral)... forward OR backward

Special Tests

- Spring Test
 - Patient prone
 - Pushing on lumbosacral junction
 - Normal response is a feeling of 'springing' down (forward) and back up
 - If it doesn't spring down/forward (anterior)... it's stuck backward (posterior)
- Sphinx Test
 - Patient prone
 - Palpate at sacral sulci and ILA
 - Patient goes onto elbows or hands → increasing lordosis and lumbosacral angle = sacral flexion
 - If dysfunction is forward, sacral flexion is going into freedom and it evens out
 - If dysfunction is backward, sacral flexion is the barrier and it gets worse

Special Tests

- Seated Flexion test
 - Thumbs on PSIS, on skin
 - Patient seated with feet flat on floor and flexes trunk forward
 - Positive on side of PSIS moving first and farthest
 - This asymmetry indicates some dysfunctional movement on that side
 - Movement happens ON an axis (around it) so axis is on opposite side of positive finding
- Dysfunctional movement...



Application

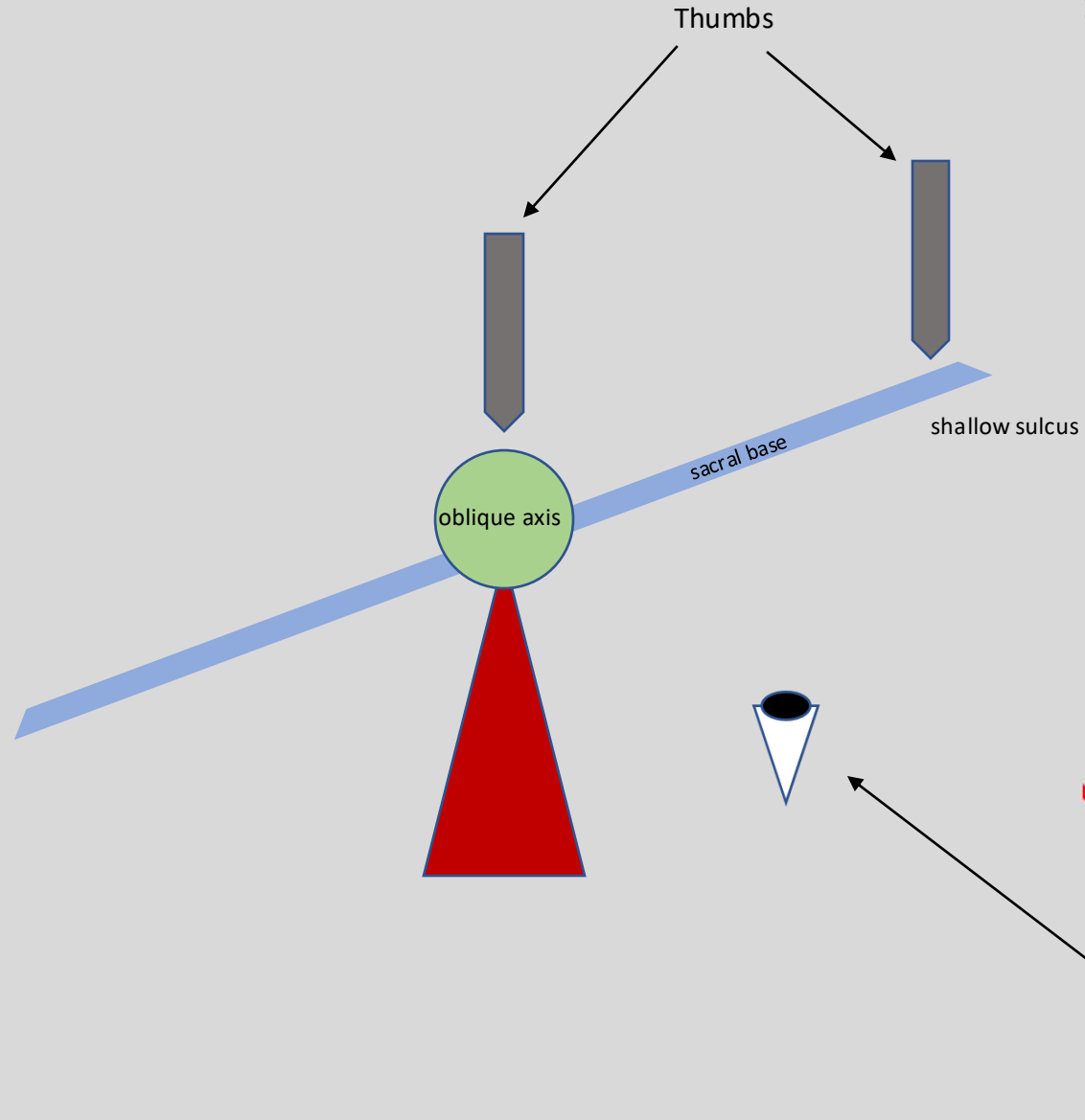
- Seated flexion test: Positive on right / Tells you...
 - Right shear or a some torsion/rotation ON a left axis)
- Spring test: Negative / Tells you...
 - (Forward unilateral flexion or forward torsion/rotation)
- Sacral sulci: Deep on right
- ILA: posterior on left
- L5 NSIRr
- Left (sacrum) ON left (axis) forward sacral torsion

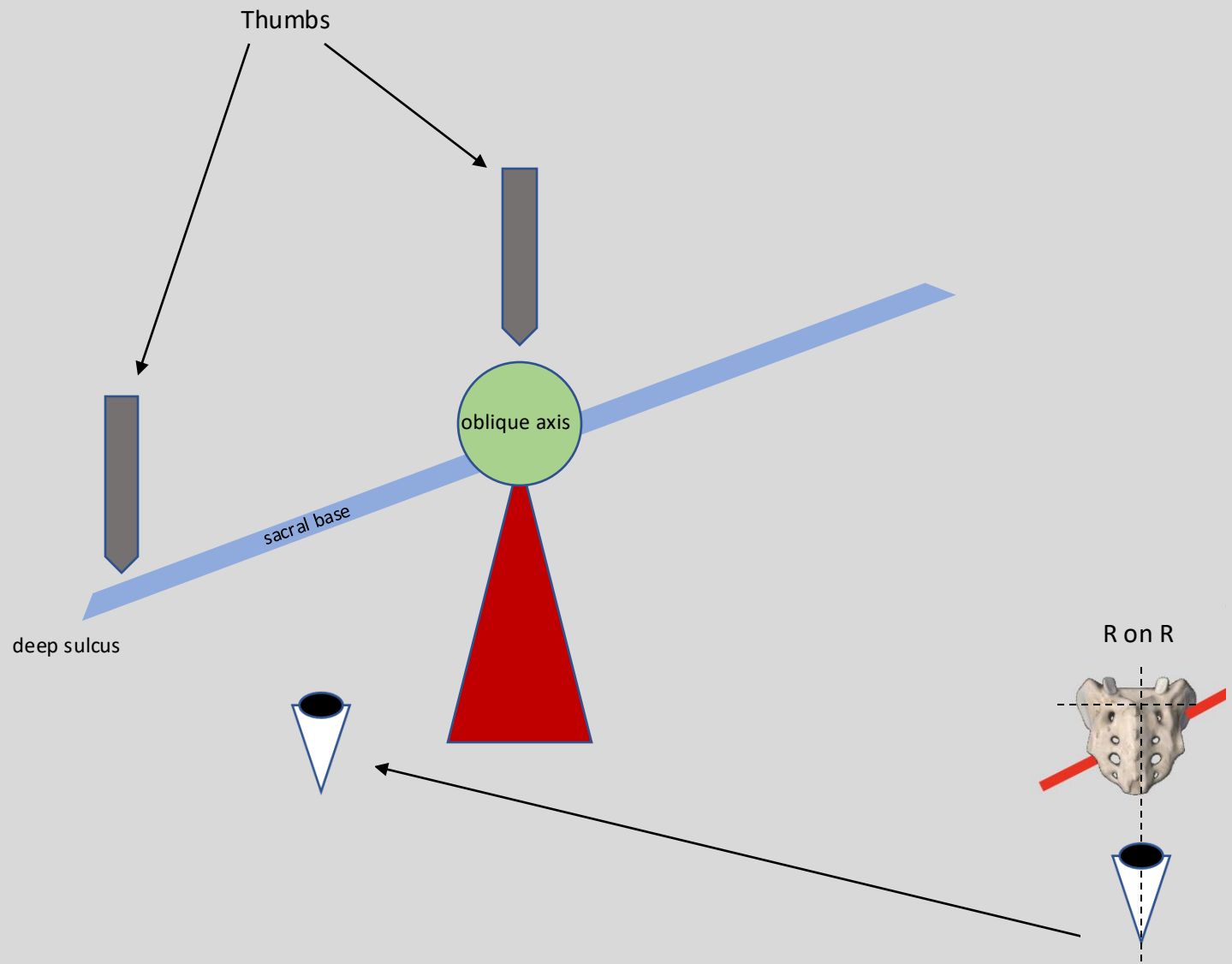
Application

- Seated flexion test: Positive on left / Tells you...
 - Left shear or torsion/rotation ON a right axis)
- Sphinx test: Positive / Tells you...
 - Backward shear or torsion/rotation
- Sacral sulci: Deep on right**
- ILA: Posterior on left**
- Left (sacrum) ON right (axis) backward sacral torsion
- All motion is relative...

Sulci and ILAs

- Motion is relative
 - A 'deep' sulcus is only deep relative to axis bearing sulcus, which feels comparatively shallow
 - A 'posterior' ILA feels posterior only because you're comparing it to the ILA on the other side, which itself feels 'anterior'
 - Your fingers only feel a difference... which finger is right?
- Know whether its forward or backward (Spring or Sphinx)
- Know where axis is (Seated flexion test, L5 position)
- Understand deep vs. shallow relativity
- Apply palpatory findings





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List of diagnoses

- R on R (right on right forward sacral torsion or rotation)
- L on L (left on left forward sacral torsion or rotation)
- R on L (right on left backward sacral torsion or rotation)
- L on R (left on right backward sacral torsion or rotation)
- LUSF/RUSF (left or right unilateral sacral flexion)
- LUSE/RUSE (left or right unilateral sacral extension)
- Bilateral sacral flexion
- Bilateral sacral extension